Signals and Systems Lecture-7: Fourier Series (1)

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Introduction (1)

Consider a periodic continuous-time signal $x(t)$ with period T_0. That is, T_0 is the smallest positive number verifying:

$$x(t) = x(t + nT_0) \ \ \forall \ \ n \in \mathbb{Z}$$

In what follows, we will see that $x(t)$ can be expanded in terms of complex exponential signals:

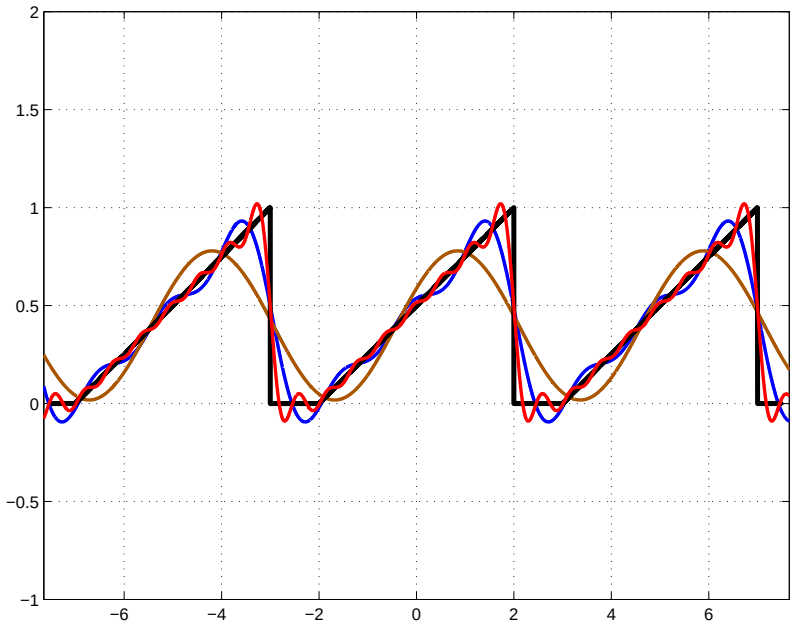
$$x(t) = \sum_{k=-\infty}^{+\infty} X[k] e^{j2\pi(kf_0)t}$$

where:

- $f_0 = \frac{1}{T_0}$ is called the fundamental frequency.
- $\{X[k]\}_{k=-\infty}^{+\infty}$ are called the Fourier Series Coefficients.
- The k-th multiple of the fundamental frequency (i.e. the frequency kf_0) is called the k-th harmonic.

Introduction (2)

Example: consider the following signal whose period is $T_0 = 5$.

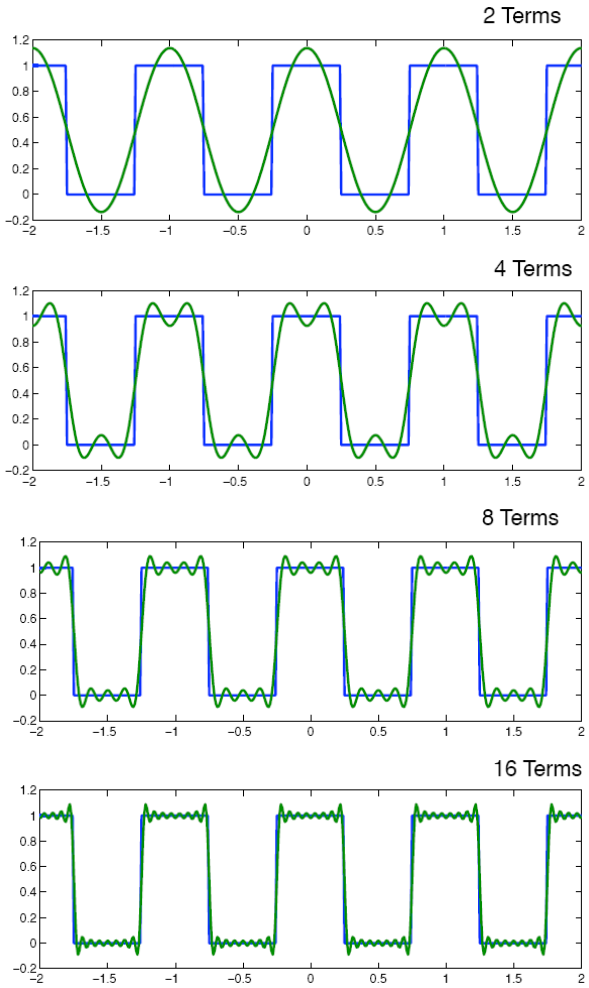


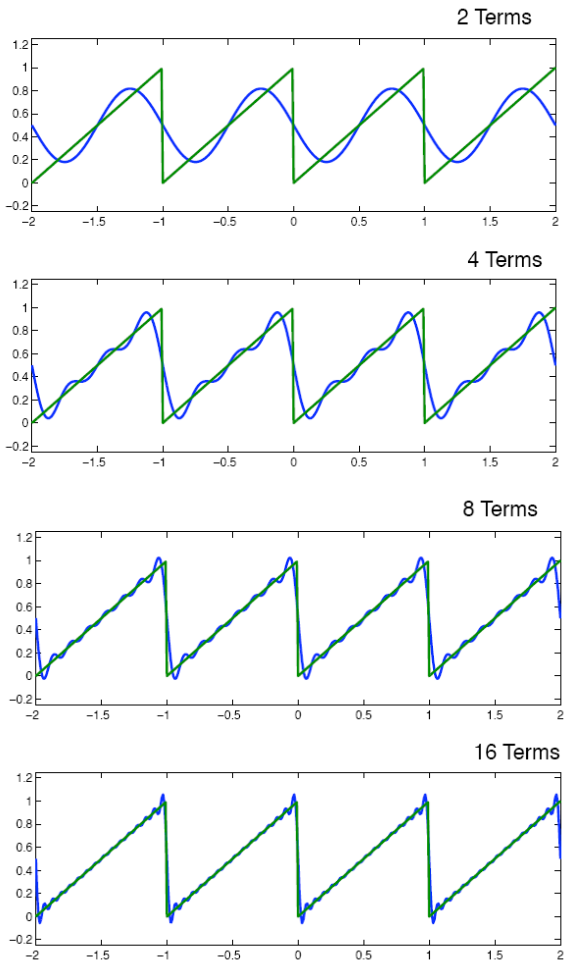
- In black, we have the signal.
- In brown, the approximation with 3 terms ($k = 0, \pm 1$).
- In blue, the approximation with 5 terms ($k = 0, \pm 1, \pm 2$).
- In red, the approximation with 7 terms ($k = 0, \pm 1, \pm 2, \pm 3$).

The signal determined from the Fourier series will approach the signal $x(t)$ when the number of harmonics is large.

Introduction (3)

- The approximation of the sawtooth and rectangular periodic signals by the Fourier series with different number of terms is shown in the figure below:

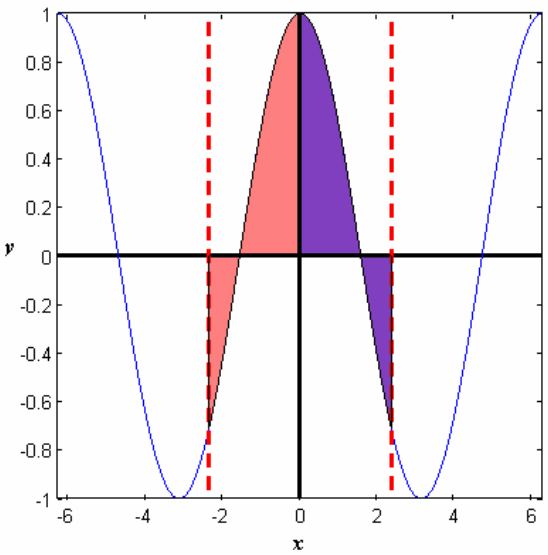




Preliminaries (1)

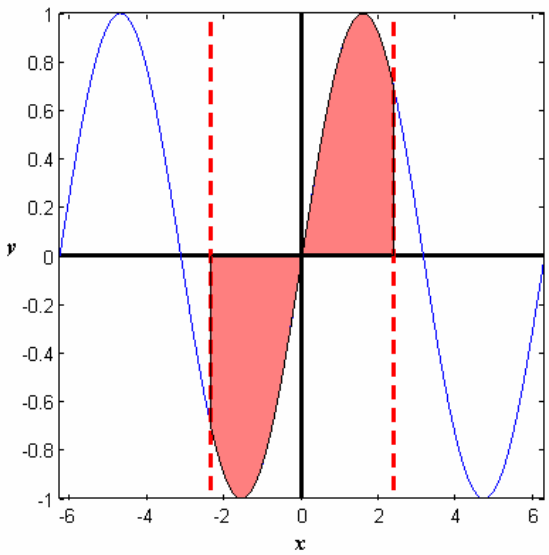
- If $f(x)$ is even then:

$$\int_{-a}^{a} f(x)\mathrm{d}x = 2\int_{0}^{a} f(x)\mathrm{d}x$$



- If $f(x)$ is odd then:

$$\int_{-a}^{a} f(x)\mathrm{d}x = 0$$



Preliminaries (2)

Consider the two functions $f_1(x)$ and $f_2(x)$ and let:

$$f(x) = f_1(x).f_2(x)$$

- If $f_1(x)$ is even and $f_2(x)$ is even, then $f(x)$ is even (**even**$\times$**even=even**):

$$f(-x) = f_1(-x)f_2(-x) = f_1(x)f_2(x) = f(x)$$

- If $f_1(x)$ is even and $f_2(x)$ is odd, then $f(x)$ is odd (**even**$\times$**odd=odd**).

$$f(-x) = f_1(-x)f_2(-x) = f_1(x)(-f_2(x)) = -f(x)$$

- If $f_1(x)$ is odd and $f_2(x)$ is odd, then $f(x)$ is even (**odd**$\times$**odd=even**).

$$f(-x) = f_1(-x)f_2(-x) = (-f_1(x))(-f_2(x)) = f(x)$$

Preliminaries (3)

- The integral of the cosine function over n periods is equal to zero (for all values of n):

$$\int_{t_0}^{t_0+nT_0} \cos(2\pi f_0 t)\,\mathrm{d}t = 0 \;\; \forall \;\; n \in \mathbb{Z}$$

  - In fact:

$$\begin{aligned}
\int_{t_0}^{t_0+nT_0} \cos(2\pi f_0 t)\,\mathrm{d}t &= \frac{1}{2\pi f_0}\left[\sin(2\pi f_0 t)\right]_{t_0}^{t_0+nT_0} \\
&= \frac{1}{2\pi f_0}\left[\sin(2\pi f_0(t_0+nT_0)) - \sin(2\pi f_0 t_0)\right] \\
&= \frac{1}{2\pi f_0}\left[\sin(2\pi f_0 t_0 + 2n\pi) - \sin(2\pi f_0 t_0)\right] \\
&= \frac{1}{2\pi f_0}\left[\sin(2\pi f_0 t_0) - \sin(2\pi f_0 t_0)\right] = 0
\end{aligned}$$

Preliminaries (4)

- The integral of the sine function over n periods is equal to zero (for all values of n):

$$\int_{t_0}^{t_0+nT_0} \sin(2\pi f_0 t)\,\mathrm{d}t = 0 \;\; \forall \;\; n \in \mathbb{Z}$$

  - In fact:

$$\begin{aligned}
\int_{t_0}^{t_0+nT_0} \sin(2\pi f_0 t)\,\mathrm{d}t &= -\frac{1}{2\pi f_0}\left[\cos(2\pi f_0 t)\right]_{t_0}^{t_0+nT_0} \\
&= -\frac{1}{2\pi f_0}\left[\cos(2\pi f_0(t_0+nT_0)) - \cos(2\pi f_0 t_0)\right] \\
&= -\frac{1}{2\pi f_0}\left[\cos(2\pi f_0 t_0 + 2n\pi) - \cos(2\pi f_0 t_0)\right] \\
&= -\frac{1}{2\pi f_0}\left[\cos(2\pi f_0 t_0) - \cos(2\pi f_0 t_0)\right] = 0
\end{aligned}$$

Preliminaries (5)

- **Definition:** Two functions $f_1(t)$ and $f_2(t)$ are orthogonal over the interval $[a \;\; b]$ if:

$$\int_a^b f_1(t) f_2^*(t) \mathrm{d}t = 0$$

where $f_2^*(t)$ stands for the complex conjugate of $f_2(t)$.
  - In other words, if $f_2(t) = f_{2r}(t) + j f_{2i}(t)$, then $f_2^*(t) = f_{2r}(t) - j f_{2i}(t)$.
  - $j = \sqrt{-1}$.
- The function $e^{j\theta}$ verifies the following relation (θ is a real number expressed in radians):

$$e^{j\theta} = \cos\theta + j\sin\theta$$
$$\Rightarrow \quad [e^{j\theta}]^* = e^{-j\theta} = \cos\theta - j\sin\theta$$

Orthogonality of the Complex Exponential functions

- **Proposition**: For $m, n \in \mathbb{Z}$:

$$I = \int_{T_0} e^{j2\pi(mf_0)t} e^{-j2\pi(nf_0)t} \mathrm{d}t = \begin{cases} T_0, & m = n; \\ 0, & m \neq n. \end{cases}$$

- *Proof:*
  - For $m = n$:

$$I = \int_{T_0} 1 \mathrm{d}t = T_0$$

  - For $m \neq n$:

$$I = \int_{T_0} e^{j2\pi(m-n)f_0 t} \mathrm{d}t = \frac{1}{j2\pi(m-n)f_0} \left[e^{j2\pi(m-n)f_0 t} \right]_{t_0}^{t_0+T_0}$$

$$= \frac{1}{j2\pi(m-n)f_0} [\underbrace{e^{j[2\pi(m-n)f_0 t_0 + 2\pi(m-n)]}}_{=e^{j2\pi(m-n)f_0 t_0}} - e^{j2\pi(m-n)f_0 t_0}] = 0$$

- This proposition implies that the functions $e^{j2\pi(mf_0)t}$ and $e^{j2\pi(nf_0)t}$ are orthogonal over any interval of duration T_0 for $m \neq n$.

Orthogonality of the sinusoidal functions (1)

- **Proposition:** For $m, n \in \mathbb{N}^*$:

$$I = \int_{T_0} \cos\left(2\pi(mf_0)t\right)\cos\left(2\pi(nf_0)t\right)\mathrm{d}t = \begin{cases} \frac{T_0}{2}, & \text{for } m = n \\ 0, & \text{for } m \neq n \end{cases}$$

- *Proof:*
  - $\cos(a)\cos(b)$ can be written as: $\cos(a)\cos(b) = \frac{1}{2}(\cos(a+b) + \cos(a-b))$.
  - Consequently, the above integral can be written as:

$$I = \frac{1}{2}\left[\int_{T_0} \cos\left(2\pi(m+n)f_0 t\right)\mathrm{d}t + \int_{T_0} \cos\left(2\pi(m-n)f_0 t\right)\mathrm{d}t\right]$$
$$= \frac{1}{2}\left[I_1 + I_2\right]$$

  - Since the integral of the cosine over any number of periods is equal to 0:

$$I_1 = \begin{cases} T_0, & m = -n; \\ 0, & m \neq -n. \end{cases} \quad ; \quad I_2 = \begin{cases} T_0, & m = n; \\ 0, & m \neq n. \end{cases}$$

Orthogonality of the sinusoidal functions (2)

- - Since m and n are nonzero natural integers, then m can not be equal to $-n$ implying that the integral I_1 is always equal to zero.
  - Consequently:

$$I = \frac{I_2}{2} = \begin{cases} T_0/2, & m = n; \\ 0, & m \neq n. \end{cases}$$

- The above proposition implies that the functions $\cos(2\pi(mf_0)t)$ and $\cos(2\pi(nf_0)t)$ are orthogonal over any interval of duration T_0 where m and n are nonzero natural integers verifying: $m \neq n$.

Orthogonality of the sinusoidal functions (3)

- **Proposition:** For $m, n \in \mathbb{N}^*$:

$$\int_{T_0} \sin\left(2\pi(mf_0)t\right) \sin\left(2\pi(nf_0)t\right) \mathrm{d}t = \begin{cases} \frac{T_0}{2}, & \text{for } m = n \\ 0, & \text{for } m \neq n \end{cases}$$

- *Proof:*
  - $\sin(a)\sin(b)$ can be written as:
    $\sin(a)\sin(b) = \frac{1}{2}(\cos(a-b) - \cos(a+b))$.
  - Consequently, the above integral can be written as:

$$I = \frac{1}{2}\left[\int_{T_0} \cos\left(2\pi(m-n)f_0 t\right) \mathrm{d}t - \int_{T_0} \cos\left(2\pi(m+n)f_0 t\right) \mathrm{d}t\right]$$
$$= \frac{1}{2}\left[I_1 - I_2\right]$$

  - Since the integral of the cosine over any number of periods is equal to 0:

$$I_1 = \begin{cases} T_0, & m = n; \\ 0, & m \neq n. \end{cases} \quad ; \quad I_2 = \begin{cases} T_0, & m = -n; \\ 0, & m \neq -n. \end{cases}$$

Orthogonality of the sinusoidal functions (4)

- - Since m and n are nonzero natural integers, then m can not be equal to $-n$ implying that the integral I_2 is always equal to zero.
  - Consequently:

$$I = \frac{I_1}{2} = \begin{cases} T_0/2, & m = n; \\ 0, & m \neq n. \end{cases}$$

- The above proposition implies that the functions $\sin(2\pi(mf_0)t)$ and $\sin(2\pi(nf_0)t)$ are orthogonal over any interval of duration T_0 where m and n are nonzero natural integers verifying: $m \neq n$.

Orthogonality of the sinusoidal functions (5)

- **Proposition:** For $m, n \in \mathbb{N}^*$:

$$\int_{T_0} \sin\left(2\pi(mf_0)t\right)\cos\left(2\pi(nf_0)t\right)\mathrm{d}t = 0 \;\; \forall \;\; m, n$$

- *Proof:*
  - $\sin(a)\cos(b)$ can be written as:
    $\sin(a)\cos(b) = \frac{1}{2}(\sin(a+b) + \sin(a-b))$.
  - Consequently, the above integral can be written as:

$$I = \frac{1}{2}\left[\int_{T_0} \sin\left(2\pi(m+n)f_0 t\right)\mathrm{d}t + \int_{T_0} \sin\left(2\pi(m-n)f_0 t\right)\mathrm{d}t\right]$$
$$= \frac{1}{2}\left[I_1 + I_2\right]$$

  - Consider the integral I_1.
    - For $m \neq -n$: The integral of the sine function over any number of periods is equal to zero.
    - For $m = -n$, $\sin\left(2\pi(m+n)f_0 t\right) = 0$ implying that $I_1 = 0$.
    - Therefore, $I_1 = 0$ for all values of m and n.

Orthogonality of the sinusoidal functions (6)

- Consider the integral I_2.
  - For $m \neq n$: The integral of the sine function over any number of periods is equal to zero.
  - For $m = n$, $\sin(2\pi(m-n)f_0 t) = 0$ implying that $I_2 = 0$.
  - Therefore, $I_2 = 0$ for all values of m and n.
- Therefore, $I = 0 \; \forall \; m, n \in \mathbb{N}^*$.
- The above proposition implies that the functions $\sin(2\pi(mf_0)t)$ and $\cos(2\pi(nf_0)t)$ are orthogonal over any interval of duration T_0 where m and n are any two nonzero natural integers.

Orthogonality of the sinusoidal functions (7)

- To summarize:

$$\int_{T_0} e^{j2\pi(mf_0)t} e^{-j2\pi(nf_0)t} \mathrm{d}t = \begin{cases} T_0, & m = n; \\ 0, & m \neq n. \end{cases}$$

$$\int_{T_0} \cos\left(2\pi(mf_0)t\right) \cos\left(2\pi(nf_0)t\right) \mathrm{d}t = \begin{cases} \frac{T_0}{2}, & \text{for } m = n \\ 0, & \text{for } m \neq n \end{cases}$$

$$\int_{T_0} \sin\left(2\pi(mf_0)t\right) \sin\left(2\pi(nf_0)t\right) \mathrm{d}t = \begin{cases} \frac{T_0}{2}, & \text{for } m = n \\ 0, & \text{for } m \neq n \end{cases}$$

$$\int_{T_0} \sin\left(2\pi(mf_0)t\right) \cos\left(2\pi(nf_0)t\right) \mathrm{d}t = 0 \ \ \forall \ \ m, n$$

  - In the first equation, $m, n \in \mathbb{Z}$.
  - In the last three equations, $m, n \in \mathbb{N}^*$.

Exponential form of the Fourier series (1)

Fourier Series representation of CT periodic signals:

- **Proposition:** A periodic CT signal $x(t)$ can be written as:

$$x(t) = \sum_{k=-\infty}^{+\infty} X[k] e^{j2\pi(kf_0)t}$$

where $f_0 = \frac{1}{T_0}$ is the fundamental frequency and $X[k]$ are the Fourier series coefficients given by:

$$X[k] = \frac{1}{T_0} \int_{T_0} x(t) e^{-j2\pi(kf_0)t} \mathrm{d}t \quad ; \quad k \in \mathbb{Z}$$

  - In general, the coefficients $X[k]$ are complex even if the signal $x(t)$ is real.

Exponential form of the Fourier series (2)

Proof:

- Consider the equation:

$$x(t) = \sum_{k=-\infty}^{+\infty} X[k] e^{j2\pi(kf_0)t}$$

- Multiplying both sides by $e^{-j2\pi(k'f_0)t}$ and integrating:

$$\int_{T_0} x(t) e^{-j2\pi(k'f_0)t} \mathrm{d}t = \sum_{k=-\infty}^{+\infty} X[k] \int_{T_0} e^{j2\pi(kf_0)t} e^{-j2\pi(k'f_0)t} \mathrm{d}t$$

$$= T_0 X[k'] + \sum_{k \neq k'} X[k] \underbrace{\int_{T_0} e^{j2\pi(kf_0)t} e^{-j2\pi(k'f_0)t} \mathrm{d}t}_{=0}$$

- Consequently:

$$X[k'] = \frac{1}{T_0} \int_{T_0} x(t) e^{-j2\pi(k'f_0)t} \mathrm{d}t$$

Exponential form of the Fourier series (3)

- **Proposition**: For real signals, the Fourier series coefficients satisfy the following relation:

$$X[-k] = X^*[k] \quad ; \quad k \in \mathbb{Z}$$

- *Proof*:
  - Since $X[k] = \frac{1}{T_0}\int_{T_0} x(t)e^{-j2\pi(kf_0)t}\mathrm{d}t$, replacing k by $-k$ results in:

$$X[-k] = \frac{1}{T_0}\int_{T_0} x(t)e^{j2\pi(kf_0)t}\mathrm{d}t$$

  - On the other hand:

$$X^*[k] = \left[\frac{1}{T_0}\int_{T_0} x(t)e^{-j2\pi(kf_0)t}\mathrm{d}t\right]^* = \frac{1}{T_0}\int_{T_0} x^*(t)\left[e^{-j2\pi(kf_0)t}\right]^*\mathrm{d}t$$

$$= \frac{1}{T_0}\int_{T_0} x(t)e^{j2\pi(kf_0)t}\mathrm{d}t$$

    - where $x^*(t) = x(t)$ since $x(t)$ is real and $\left(e^{-j\theta}\right)^* = e^{j\theta}$.
  - Consequently, $X[-k] = X^*[k]$.

Trigonometric continuous-time Fourier series (1)

- For **real** periodic continuous-time signals, an alternative representation of the Fourier series is given by:

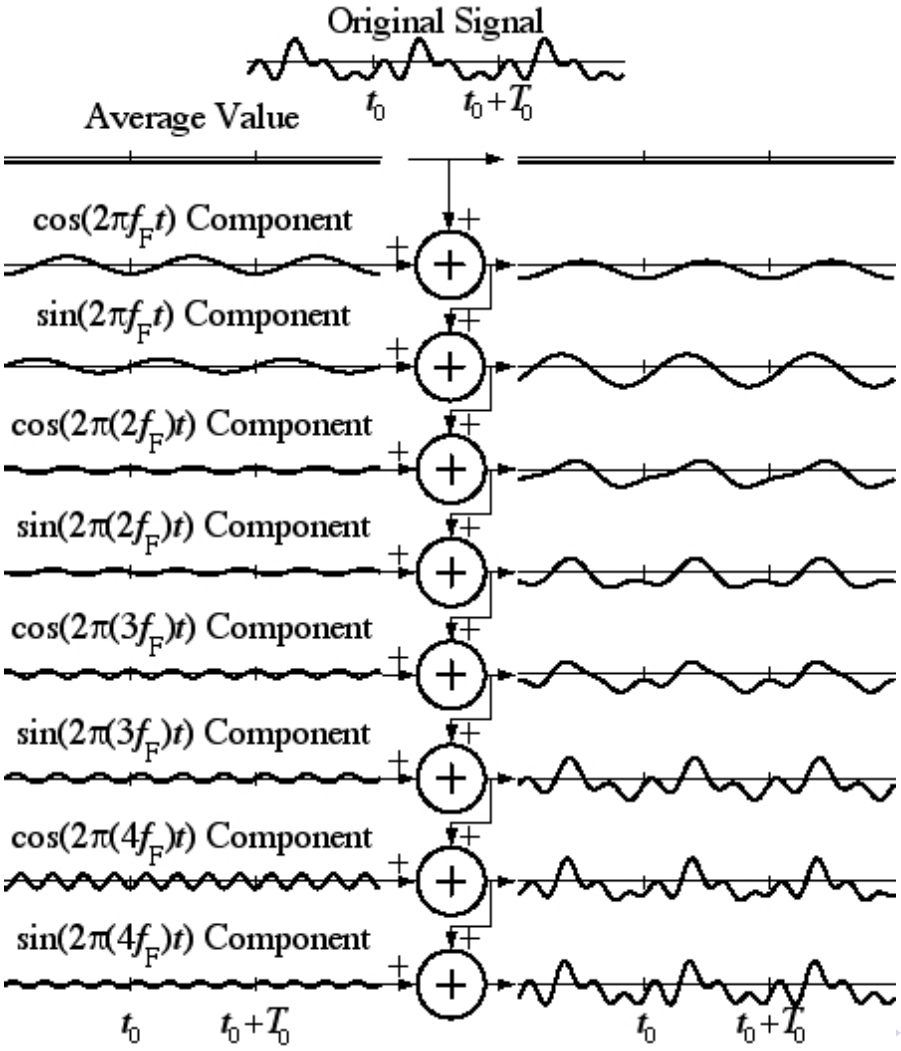
$$x(t) = X_c[0] + \sum_{k=1}^{+\infty} \{X_c[k]\cos(2\pi(kf_0)t) + X_s[k]\sin(2\pi(kf_0)t)\}$$

- where:

$$\begin{aligned} X_c[0] &= \frac{1}{T_0}\int_{T_0} x(t)\mathrm{d}t \\ X_c[k] &= \frac{2}{T_0}\int_{T_0} x(t)\cos(2\pi(kf_0)t)\mathrm{d}t \\ X_s[k] &= \frac{2}{T_0}\int_{T_0} x(t)\sin(2\pi(kf_0)t)\mathrm{d}t \end{aligned}$$

Trigonometric continuous-time Fourier series (2)

- The following figure illustrates the concept of representing a periodic signal as a linear combination of sinusoids ($f_F = f_0 = 1/T_0$ where T_0 is the period of the signal):



Trigonometric continuous-time Fourier series (3)

- The Fourier series follows from the orthogonality properties of the sinusoidal functions. For example, let's determine the values of $X_c[k]$ for $k \neq 0$. In fact:

$$x(t) = X_c[0] + \sum_{k=1}^{+\infty} \{X_c[k]\cos(2\pi(kf_0)t) + X_s[k]\sin(2\pi(kf_0)t)\}$$

- Multiplying both sides by $\cos(2\pi(k'f_0)t)$:

$$\begin{aligned} x(t)\cos(2\pi(k'f_0)t) &= X_c[0]\cos(2\pi(k'f_0)t) \\ &+ \sum_{k=1}^{+\infty} X_c[k]\cos(2\pi(kf_0)t)\cos(2\pi(k'f_0)t) \\ &+ \sum_{k=1}^{+\infty} X_s[k]\sin(2\pi(kf_0)t)\cos(2\pi(k'f_0)t) \end{aligned}$$

Trigonometric continuous-time Fourier series (4)

- Integrating both sides:

$$\int_{T_0} x(t)\cos(2\pi(k'f_0)t)\mathrm{d}t = X_c[0]\underbrace{\int_T \cos(2\pi(k'f_0)t)\mathrm{d}t}_{=0\ \ \forall\ \ k'}$$

$$+\sum_{k=1}^{+\infty} X_c[k]\underbrace{\int_T \cos(2\pi(kf_0)t)\cos(2\pi(k'f_0)t)\mathrm{d}t}_{=\begin{cases} 0, & k\neq k'; \\ \frac{T_0}{2}, & k=k'. \end{cases}}$$

$$+\sum_{k=1}^{+\infty} X_s[k]\underbrace{\int_T \sin(2\pi(kf_0)t)\cos(2\pi(k'f_0)t)\mathrm{d}t}_{=0\ \ \forall\ \ k,k'}$$

- Therefore:

$$\int_{T_0} x(t)\cos(2\pi(k'f_0)t)\mathrm{d}t = X_c[k']\frac{T_0}{2} \Rightarrow X_c[k'] = \frac{2}{T_0}\int_{T_0} x(t)\cos(2\pi(k'f_0)t)\mathrm{d}t$$

The Trigonometric Fourier series: even functions

Proposition: For even functions:

$$X_s[k] = 0 \quad \forall \quad k$$

Proof:

- $X_s[k]$ can be written as:

$$X_s[k] = \frac{2}{T_0} \int_{T_0} x(t) \sin\left(2\pi(kf_0)t\right) \mathrm{d}t$$
$$= \frac{2}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t) \sin\left(2\pi(kf_0)t\right) \mathrm{d}t$$

  - $x(t)$ is even.
  - $\sin\left(2\pi(kf_0)t\right)$ is odd.
  - Consequently: $x(t) \sin\left(2\pi(kf_0)t\right)$ is odd and the last integral is equal to zero.

The Trigonometric Fourier series: odd functions

- **Proposition:** For odd functions:

$$X_c[k] = 0 \quad \forall \quad k$$

- *Proof:*
  - For $k = 0$:

$$X_c[0] = \frac{1}{T_0}\int_{T_0} x(t)\mathrm{d}t = \frac{1}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t)\mathrm{d}t = 0$$

    since $x(t)$ is odd.
  - For $k \neq 0$:

$$X_c[k] = \frac{2}{T_0}\int_{T_0} x(t)\cos\left(2\pi(kf_0)t\right)\mathrm{d}t = \frac{2}{T_0}\int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t)\cos\left(2\pi(kf_0)t\right)\mathrm{d}t$$

    - Since $x(t)$ is odd and $\cos\left(2\pi(kf_0)t\right)$ is even, then $x(t)\cos\left(2\pi(kf_0)t\right)$ is odd and the last integral is equal to zero.

Exponential and Trigonometric forms (1)

Proposition: For real signals, the exponential Fourier series coefficients are related to the trigonometric coefficients by:

$$X[0] = X_c[0]$$
$$X[k] = \frac{X_c[k] - jX_s[k]}{2} \quad ; \quad k = 1, 2, \ldots$$

Proof:

- The exponential Fourier Series (FS) coefficients $X[k]$ are given by:

$$X[k] = \frac{1}{T_0} \int_{T_0} x(t) e^{-j2\pi(kf_0)t} \mathrm{d}t$$

  - For $k = 0$:

$$X[0] = \underbrace{\frac{1}{T_0} \int_{T_0} x(t) \mathrm{d}t}_{=X_c[0]} = X_c[0]$$

Exponential and Trigonometric forms (2)

- For $k > 0$ (since $e^{-j\theta} = \cos\theta - j\sin\theta$):

$$
\begin{aligned}
X[k] &= \frac{1}{T_0}\int_{T_0} x(t)\left[\cos(2\pi(kf_0)t) - j\sin(2\pi(kf_0)t)\right]\mathrm{d}t \\
&= \frac{1}{2}\underbrace{\frac{2}{T_0}\int_{T_0} x(t)\cos(2\pi(kf_0)t)\mathrm{d}t}_{=X_c[k]} - j\frac{1}{2}\underbrace{\frac{2}{T_0}\int_{T_0} x(t)\sin(2\pi(kf_0)t)\mathrm{d}t}_{=X_s[k]} \\
&= \frac{X_c[k] - jX_s[k]}{2}
\end{aligned}
$$

Exponential and Trigonometric forms (3)

- To summarize, for real signals, the exponential FS coefficients are related to the trigonometric FS coefficients by:

$$
\begin{aligned}
X[0] &= X_c[0] \\
X[k] &= \frac{X_c[k] - jX_s[k]}{2} \quad k = 1, 2, \ldots \\
X[-k] &= X^*[k] = \frac{X_c[k] + jX_s[k]}{2} \quad k = 1, 2, \ldots
\end{aligned}
$$

- In an equivalent way, the trigonometric FS coefficients are related to the exponential FS coefficients by:

$$
\begin{aligned}
X_c[0] &= X[0] \\
X_s[0] &= 0 \\
X_c[k] &= 2\mathrm{Re}\{X[k]\} \quad k = 1, 2, \ldots \\
X_s[k] &= -2\mathrm{Im}\{X[k]\} \quad k = 1, 2, \ldots
\end{aligned}
$$

Exponential and Trigonometric forms (4)

- The trigonometric FS coefficients are defined only when the CT signal $x(t)$ is real.
  - The exponential FS coefficients $X[k]$ are defined for $k \in \mathbb{Z}$. They are complex-valued.
  - The trigonometric FS coefficients $X_c[k]$ are defined for $k \in \mathbb{N}$. They are real-valued.
  - The trigonometric FS coefficients $X_s[k]$ are defined for $k \in \mathbb{N}^*$. They are real-valued.
- For real signals $x(t)$: $X[-k] = X^*[k]$.
  - For real and even signals: $X_s[k] = 0$ implying that $X[k]$ is real.
  - For real and odd signals: $X_c[k] = 0$ implying that $X[k]$ is pure imaginary.

Parseval's Theorem (1)

Parseval's Theorem:

- The average power of a periodic CT signal $x(t)$ is related to is exponential FS coefficients by:

$$P_{av} = \frac{1}{T_0}\int_{T_0} |x(t)|^2 \mathrm{d}t = \sum_{k=-\infty}^{+\infty} |X[k]|^2$$

- If $x(t)$ is real:

$$\begin{aligned} P_{av} &= |X[0]|^2 + \sum_{k=-\infty}^{-1} |X[k]|^2 + \sum_{k=1}^{+\infty} |X[k]|^2 \\ &= |X[0]|^2 + \sum_{k=1}^{+\infty} \underbrace{|X[-k]|^2}_{=|X^*[k]|^2=|X[k]|^2} + \sum_{k=1}^{+\infty} |X[k]|^2 \\ &= |X[0]|^2 + 2\sum_{k=1}^{+\infty} |X[k]|^2 \end{aligned}$$

Parseval's Theorem (2)

- The DC power of the signal is equal to:

$$P_0 = |X[0]|^2 \quad \mathrm{W}$$

- The power of the k-th harmonic of a real signal is equal to:

$$P_k = |X[k]|^2 + |X[-k]|^2 = 2|X[k]|^2 \quad \mathrm{W}$$

since $X[-k] = X^*[k]$.

- The average value of the signal is given by:

$$x_{\mathrm{av}} = \frac{1}{T_0}\int_{T_0} x(t)\mathrm{d}t = X[0]$$